

Vidhi Dixit

Software Engineer — C/C++ · Python · React.js · AI · Full-Stack Systems

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PROFESSIONAL SUMMARY

Software Engineer with around 3 years of experience across full-stack web development and systems-level programming in C/C++ and Python. Skilled in building secure RESTful APIs and microservices with FastAPI, multithreaded and performance-oriented C/C++ services, and PostgreSQL database design. Experienced with Docker containerization, AWS cloud deployment, GitHub Actions CI/CD, and Agile development. Independently designed and built Supra Engine Analytics, a FastAPI and React platform that computes a user's accessible clinical knowledge set through a graph traversal and permission-filtering pipeline, with an interactive analytics dashboard over the results. Also developed and deployed a Retrieval-Augmented Generation (RAG) based AI agent on AWS, applying LLM APIs, vector databases, and containerized cloud infrastructure. Comfortable working across Windows and Linux environments and quick to pick up new languages, tools, and domains.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, JavaScript, TypeScript
- **Frameworks:** FastAPI, React.js, Next.js, Vite, SQLAlchemy, LWC, Apex
- **Architecture:** Microservices, RESTful API Design, System Design Fundamentals
- **Systems & Performance:** Multithreading, concurrency, memory/resource management, graph traversal algorithms (BFS), performance profiling and optimization
- **Databases:** PostgreSQL, Supabase, SQL, SQLite
- **DevOps & Cloud:** Docker, Git, CI/CD (GitHub Actions), AWS (EC2)
- **Operating Systems:** Windows 10/11, Windows Server, Linux
- **Tools & IDEs:** VS Code, Visual Studio, Postman, Swagger/OpenAPI, GitHub, Bitbucket
- **Frontend & Data Visualization:** React Flow, Recharts, Material UI, Axios, responsive dashboard design
- **AI Engineering:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, AI API Integration, AI Agent Development, Cloud Deployment of AI Services, Intelligent Workflow Automation
- **Software Engineering:** OOP, Data Structures & Algorithms, SDLC, Agile/Scrum, Code Review, Debugging, Secure Coding Practices
- **Testing:** Unit Testing, API Testing, System & Integration Testing
- **CRM Technologies:** Salesforce Data Modeling, Workflow Automation, Requirement Analysis

WORK EXPERIENCE

Technical Consultant / Software Developer — Blue Flame Labs

May 2025 – Current

- Designed, developed, tested, and maintained scalable React.js and FastAPI applications following component-based architecture and complete SDLC practices.
- Built and maintained backend services using a microservices architecture with FastAPI and Docker, enabling independent deployment, scalability, and clearer separation of concerns across application modules.
- Designed and implemented secure RESTful APIs with authentication, authorization, request validation, exception handling, pagination, and third-party API integrations.
- Developed reusable React components, custom hooks, and responsive admin dashboards to improve maintainability and user experience.
- Designed normalized PostgreSQL database schemas and optimized SQL queries, improving backend performance and efficient high-volume data retrieval.
- Containerized frontend and backend services using Docker and supported AWS cloud deployments through GitHub Actions-based CI/CD workflows.
- Participated in Agile sprint planning, Git-based version control, pull requests, code reviews, debugging, testing, and production support.
- Maintained REST API documentation using Swagger/OpenAPI and followed software engineering best practices for scalable application development.
- Owned end-to-end ticket lifecycle — from requirement gathering and business stakeholder discussions through development, testing, deployment, and production support — ensuring delivery stayed aligned with business objectives.
- Partnered directly with business stakeholders to translate business processes and requirements into technical specifications, bringing product and process context into sprint planning and technical decisions.
- Led end-to-end testing across API and integration layers, identifying performance bottlenecks and driving optimizations that enhanced application responsiveness and reliability.

Software Developer — Cloud Synapps

June 2023 – January 2025

- Designed and developed a secure authentication and user management service using FastAPI and Python, implementing JWT-based authentication, role-based access control (RBAC), and password hashing for secure user access.
- Built scalable authentication APIs supporting user registration, login, profile management, and centralized identity management across multiple applications.

- Implemented secure OTP-based password reset and email verification workflows, improving account recovery reliability and reducing authentication-related support issues by approximately 30%.
- Designed modular and reusable API architecture with robust request validation, exception handling, and clean code practices, improving maintainability and reducing development effort for new features.
- Optimized API and database performance through efficient query design and backend optimizations, reducing average response time by approximately 25%.
- Optimized SOQL queries and handled Governor Limits, improving transaction efficiency.
- Collaborated in feature development, debugging, code reviews, testing, and application maintenance while following Agile methodologies, Git-based version control, and REST API best practices.
- Managed the complete ticket lifecycle end-to-end, from business requirement discussions and technical analysis through development, QA, deployment, and post-release support, ensuring solutions met real business needs.
- Engaged with business teams to understand authentication and identity workflows across client applications, aligning technical architecture with operational and compliance requirements.
- Drove end-to-end testing — unit, integration, and regression — for authentication services, proactively identifying performance issues and implementing fixes that improved system reliability.
- Took initiative in proposing process and workflow improvements during team discussions, contributing to more predictable sprint delivery.

PROJECTS (INDEPENDENT / SELF-DIRECTED)

Supra Engine Analytics — FastAPI, React, Vite, PostgreSQL (Supabase), SQLAlchemy, React Flow, Recharts

- Built a full-stack analytics platform that computes a user's accessible clinical knowledge set by traversing a healthcare knowledge graph and applying a sequential permission-filtering pipeline.
- Implemented breadth-first search (BFS) traversal from a user-specific entry point, Zone 2 node injection, and five sequential permission checks to produce the final candidate set.
- Designed a permission compiler providing O(1) permission lookups, removing repeated evaluation cost from the filtering stage.
- Developed the backend in FastAPI with SQLAlchemy over PostgreSQL hosted on Supabase, exposing the pipeline through documented REST endpoints with Swagger/OpenAPI.
- Built an interactive React dashboard with hierarchy (DAG) visualization, a filter funnel showing attrition at each permission stage, per-stage pipeline timing metrics, and a multi-user comparison view.
- Structured the repository as separate backend and frontend applications with environment-based configuration and versioned database schema and seed data.

AI Agent on AWS — RAG, LLM APIs, Vector Database, FastAPI, Docker, AWS EC2

- Developed and deployed a Retrieval-Augmented Generation (RAG) based conversational AI agent that answers questions about a person's background, skills, and experience on their behalf — designed to be reusable so other users can configure it with their own profile data.
- Implemented the retrieval pipeline: document ingestion, chunking, embedding generation, and storage in a vector database, paired with prompt orchestration against an LLM API to generate grounded, context-aware responses.
- Built the agent backend in FastAPI and deployed it on AWS, containerizing the service with Docker and running it on EC2 with environment-based configuration and secure handling of API credentials.
- Set up the AWS deployment end to end — instance provisioning, security groups, and networking — to expose the agent as a publicly accessible, always-on service.

EDUCATION

B.Tech in Computer Science — Sagar Institute of Research and Technology, Bhopal, India

Aug 2019 – June 2023

- Relevant Coursework: Object-Oriented Programming, Databases, Data Structures and Algorithms, Operating Systems, Computer Networks, Software Engineering.

CERTIFICATIONS

- Salesforce Certified Platform Developer I
- Salesforce Certified Advanced Administrator
- Salesforce Certified OmniStudio Developer